

Implementing the pool bias

What circuitry Model IV would need, and what the anatomy allows

2026-08-20 · companion to bias_locus.pdf

The question

bias_locus.pdf settled which model we run: Model IV, the bias into the pool. The pool neuron integrates the attention-weighted responses of its point-set,

$$\tau_S \dot{S} = -S + \sum_{\theta} (1 + a_{\theta}) R_{\theta}$$

while the cells' own drive carries no bias at all. Read literally that says each V1 cell's synapse onto the pool is individually weighted by a direction-tuned top-down signal. That is a strong anatomical claim, and it is the one GS questioned: it would need a synapse type that may not exist.

Three further things make the literal reading hard. The weighting must be PATHWAY-SPECIFIC, because the same V1 cell also projects to MT and V4 and Model IV needs that copy left alone. It must be SYNAPSE-SPECIFIC, picking out one terminal among those converging on a single pool neuron. And it must be RE-ADDRESSABLE, since the attended direction changes from trial to trial.

Four ways it could be built

implementation	cells / point-set	what it needs
Model III (for comparison)	17	tuned feedback onto V1 dendrites
IV, axo-axonic on terminals	17	a synapse type not evidenced in cortex
IV, match-cell bank	25	a parallel tuned population
IV, dendritic gating of the pool cell	17	feedback onto the POOL cell's dendrites

Counts are per point-set for motion plus colour, with one pool neuron shared across attributes; multiply by 121 for the locked grid. The recommendation is the last row, and the rest of this note is the evidence for it.

What the anatomy says

Feedback lands on dendrites, and acts branch by branch

- [1]** Cortico-cortical feedback engages active dendrites in visual cortex.
Nature 2023 (LM to V1, mouse; all-optical connectivity mapping).
[nature.com/articles/s41586-023-06007-6](https://www.nature.com/articles/s41586-023-06007-6)

"A substantial proportion of feedback inputs innervate pyramidal cell apical dendrites in layer 1." Individual feedback-recipient SPINES were identified on apical tufts. The effect is branch-specific: stimulating the feedback input "preferentially enhanced calcium signals in the branch containing the activated spine relative to the reference branch." There is no mention of feedback contacting axons or axon terminals.

Cortical axo-axonic synapses exist, but gate the WHOLE output

- [2]** Theory of axo-axonic inhibition. PLOS Comput Biol 2025.
Specific and comprehensive genetic targeting reveals brain-wide distribution and synaptic input patterns of GABAergic axo-axonic interneurons. 2024.

The axon initial segment "harbors GABA-A receptors contacted by interneurons, most of which are axo-axonic (chandelier) cells" -- about 60% of the synapses on the proximal axon of mouse visual cortical cells. They are GABAergic, local, and control ACTION POTENTIAL INITIATION. That is decisive against the literal reading: modulating the initial segment gates everything the cell sends, so it cannot weight the pool's copy while sparing the MT / V4 copy.

Presynaptic modulation of cortical terminals is real but diffuse

- [3]** Pre- and postsynaptic activation of GABA-B receptors modulates principal cell excitation in the piriform cortex. 2018.

Presynaptic GABA-B receptors do sit on intracortical glutamatergic terminals and do reduce excitatory input. But they are engaged HETEROSYNAPTICALLY, by spillover -- diffuse, and inhibitory. Nothing here is a tuned, terminal-selective, trial-by-trial re-addressable projection.

Dendrites order their inputs by tuning

- [4]** Wilson, Whitney, Scholl & Fitzpatrick. Orientation selectivity and the functional clustering of synaptic inputs in primary visual cortex. Nat Neurosci 2016.
[nature.com/articles/nn.4323](https://www.nature.com/articles/nn.4323)

Dendritic spines cluster according to orientation preference; the degree of clustering on a single neuron PREDICTS its somatic orientation selectivity, and branches carrying co-tuned clusters show more local dendritic calcium events. So a dendritic tree can carry its afferents in tuning order.

The implementation the anatomy allows

Put [1] and [4] together. Model IV needs the pool's copy of the response weighted and the read-out's copy untouched. PATHWAY SPECIFICITY IS FREE ON THE POSTSYNAPTIC SIDE: modulate the pool neuron's dendrite and the V1 cell is never touched at all, so its projection to MT and V4 is unaffected by construction.

If the V1-to-pool synapses are arranged on the pool cell's tree in tuning order -- which [4] says dendrites do -- then an ordinary layer-1 feedback axon running through that tree, making en-passant synapses branch by branch with strength proportional to the attention profile, computes

$$\sum_{\theta} a_{\theta} R_{\theta}$$

which added to the plain sum the pool already takes gives exactly the Model IV pool input. No axo-axonic synapse. No parallel population. No extra neurons, and the synapse type is the one [1] documents, branch specificity included.

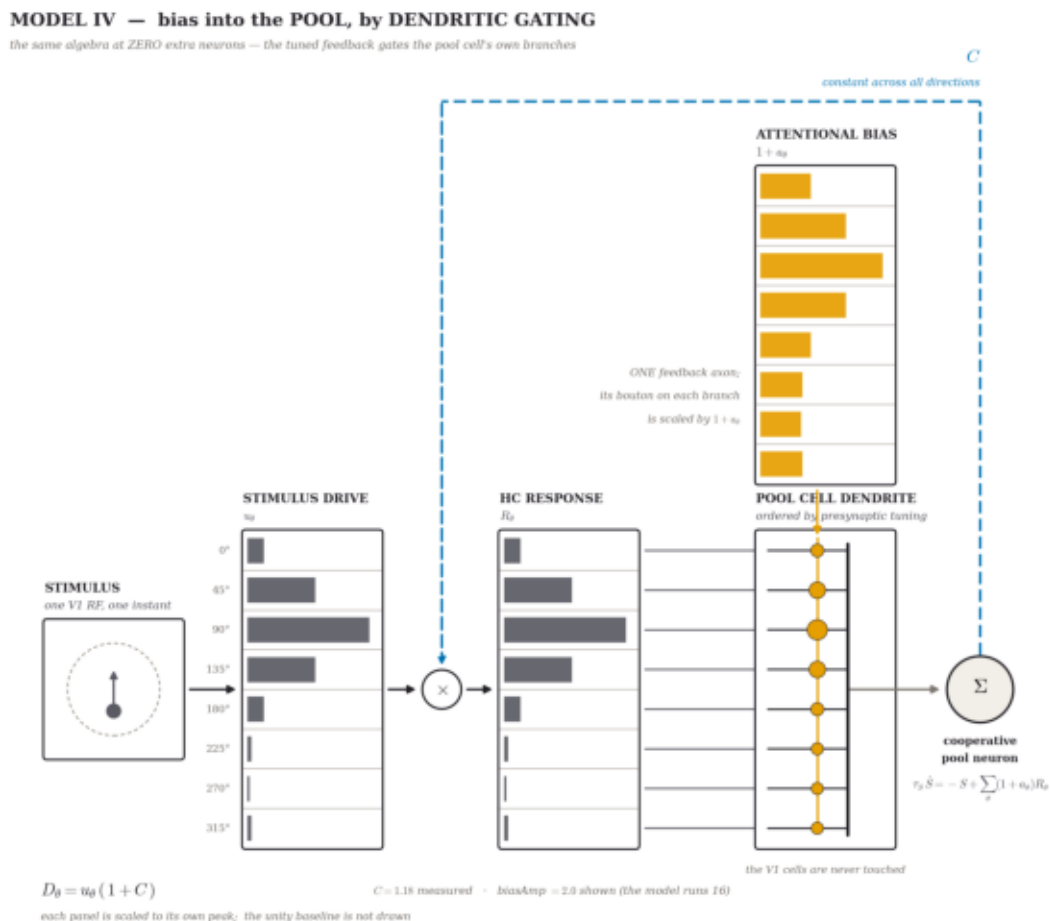


Figure: fig_modelIVD_profile.png, from ToyModel/fig_bias_profile.py IVD. No attention reaches the gain at all: the gain is the single cooperative scalar C , which is why it arrives on the spine as a bare line and gets no panel. The tuned feedback axon instead runs down the pool cell's dendritic field, its bouton on each branch scaled by the model's own bias profile, so the branch hands $(1+a)R$ to the trunk and the V1 cells are never touched.

What this rests on, and what is NOT established

The NEGATIVE half of the argument is well supported. [2] and [3] between them say that the only cortical machinery for modulating a cell's axonal output either gates the entire output (the initial segment) or is diffuse and inhibitory (presynaptic receptors on terminals). Nothing found supports a tuned, terminal-selective, trial-by-trial re-addressable modulation, so the literal reading of Model IV should be set aside.

The POSITIVE proposal is SUGGESTIVE, not established. It rests on two steps that the cited work does not actually demonstrate:

(a) that the pool cell's afferents are arranged on its tree IN TUNING ORDER. [4] shows spines CLUSTER by orientation preference and that the degree of clustering predicts somatic selectivity. Clustering is not the same thing as a systematic one-channel-per-branch map, which is what the figure draws. On this point the figure is schematic, and deliberately so.

(b) that feedback can select branches BY THE TUNING OF THE INPUTS THEY CARRY. [1] gives branch-specific feedback effects, but its organising variable is RETINOTOPIC rather than featural -- so on its own evidence it does not supply feature-selective branch targeting.

This is the weakest link in the proposal, and it is the same assumption flagged overleaf.

Two further limits worth stating plainly. The cooperative pool neuron is a MODEL CONSTRUCT: no cell type has been identified with its properties. And the search behind this note was a handful of targeted queries on 2026-08-20, not a systematic review -- primate EM work on feedback synaptic targets was not retrieved, and nothing here rests on it.

Two further findings, and they change the ranking

Tuned wiring is not required — and that cuts both ways

[5] Near-random connections support top-down feature-based attentional modulations in early sensory cortex. PLOS Computational Biology, 2025.
journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1013396

Tuning-matched projections are NOT necessary for feature-specific modulation. Near-random feedback suffices, because "the randomness of the feedback connections ensures that incidental feedback signals largely cancel out" -- spurious cross-talk destructively interferes while the useful signal survives. The authors add that "a delicate balance between randomness and structure" best reproduces the spatial globality of feature-based attention, and that TOO MUCH structure is actively harmful, producing spurious representations in unstimulated regions.

This dissolves the addressing problem -- how would a tuned axon find the theta-preferring cells? -- and with it what looked like Model III's main cost, sorting 968 V1 cells by tuning. But it cuts AGAINST dendritic gating. Near-random connectivity is a good account of "across many cells, statistically the right ones get more" and a poor account of "within one cell, the right branch". Sub-cellular addressing is precisely what randomness cannot supply.

under near-random feedback

Model III	HELPED — documented synapse type, and no addressing needed
Model IV, match-cell bank	helped — a tuned population selects statistically
Model IV, dendritic gating	WEAKENED — needs the one thing randomness cannot give

So the recommendation on page 3 is downgraded. If Model IV is to be built, a small tuned population is more compatible with unstructured feedback than branch-level gating is: the extra cells buy robustness to random wiring. Model III needs neither.

A direct prefrontal projection does carry attentional gain

[6] Hüer, Saxena & Treue. Pathway-selective optogenetics reveals the functional anatomy of top-down attentional modulation in the macaque visual cortex. PNAS 121(3), 2024.

Optogenetic inhibition of the FEF-to-MT axons, in MT, cut attentional modulation by about a third (median 19.4% to 13.6%) with NO significant effect on firing rates overall -- and push-pull: attend-in fell 2.1%, attend-out rose 3.4%. A direct prefrontal-to-visual projection therefore carries a substantial part of attentional modulation and does so without changing the driven response, which is the signature both our models assume of a pure gain term.

Two cautions. It is a SPATIAL attention task: the FEF-to-MT projection is "assumed to be retinotopically organized" and the paper does not test whether the input is tuned to the neurons' direction preference. So it does not supply the feature-tuned bias either. And the pathway selectivity is the EXPERIMENTER'S, achieved by expressing opsins in the axonal membrane -- it is not evidence that the brain modulates selected terminals endogenously, and must not be read as support for the axo-axonic route.

What this costs, and what it does not settle

The dendritic version costs exactly what Model III costs -- 17 neurons per point-set, 2057 on the locked grid -- so the choice between the two routes is no longer a choice about how much machinery the brain must find. That reverses the earlier reading, on which Model IV looked roughly one and a half times as expensive. What separates them now is only WHERE the same tuned feedback lands: on the V1 cells' dendrites (III), or on the pool cell's (IV).

An assumption BOTH models make, and [1] does not support

The feedback in [1] is organised RETINOTOPICALLY, not by feature: a relatively suppressive centre and a relatively facilitating surround, indexed by position in visual space rather than by the stimulus. Model III and Model IV both assume a DIRECTION-TUNED top-down signal, and that study does not supply one. It is mouse LM-to-V1 and a single paper, and primate feature-similarity gain implies feature-specific top-down signals exist somewhere -- but the tuned bias should be carried as an assumption, not as an established fact.

The physiological test, already measured in the model

Whichever implementation is used, the two routes differ in a way single units can see. Cued/uncued response ratio per direction channel, within one point-set:

	within one point-set	pooled across point-sets
Model III	spread ~28,000x	27,026x
Model IV	median 1.03	1.84x

Model III suppresses the non-preferred channels to 0.006 of their uncued rate while boosting the preferred one 159-fold; Model IV multiplies the whole column by one number. Near-uniform modulation across a hypercolumn versus strong tuning is the cleanest discriminator we have, and unlike the colour index it does not depend on the read-out stage. Scripts: hcps_modtuning_check.m, hcps_modtuning_percol.m.

Model IV's within-column modulation is not EXACTLY flat -- median spread 1.031 rather than 1.000 -- and that is the cells' own leaky integration, not a breach of the one-scalar structure: R lags D, so channels whose drive has a different time course integrate different stretches of the gain trajectory. Confirmed by ablation (hcps_flatness_ablate.m): at tau 0.15 the spread is 1.000000 to all printed digits, while shrinking the pool's tau_E instead barely moves it.

References

- [1] Cortico-cortical feedback engages active dendrites in visual cortex. Nature, 2023.
- [2] Theory of axo-axonic inhibition. PLOS Computational Biology, 2025. · Genetic targeting of GABAergic axo-axonic interneurons: brain-wide distribution, 2024.
- [3] Pre- and postsynaptic activation of GABA-B receptors modulates principal cell excitation in the piriform cortex, 2018.
- [4] Wilson, Whitney, Scholl & Fitzpatrick. Orientation selectivity and the functional clustering of synaptic inputs in primary visual cortex. Nature Neuroscience, 2016.
- [5] Near-random connections support top-down feature-based attentional modulations in early sensory cortex. PLOS Computational Biology, 2025.
- [6] Hürer, Saxena & Treue. Pathway-selective optogenetics reveals the functional anatomy of top-down attentional modulation in the macaque visual cortex. PNAS, 2024.

Searched 2026-08-20; full notes with URLs in the project memory file feedback-synapse-targets-lit.md. These are the sources the claims above rest on -- anything not attributed here is model measurement, not literature.